

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283356

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KIMURA; Masaru et al.

ELECTRIFIED VEHICLE

Abstract

The disclosed electrified vehicle includes at least one replaceable battery, an electric motor that exchanges electric power with the battery, a battery case, and a locking device, wherein the battery case includes a case body having an opening through which the battery is inserted and removed, and a lid that opens and closes the opening of the case body, and wherein the locking device locks the lid of the battery case in a non-openable manner when at least an electrified vehicle start-up switch is turned on.

Inventors: KIMURA; Masaru (Toyota-shi, JP), OHNISHI; Kohei (Toyota-shi, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Family ID: 96929749

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Appl. No.: 18/976906

Filed: December 11, 2024

Foreign Application Priority Data

JP 2024-034996 Mar. 07, 2024

Publication Classification

Int. Cl.: E05B81/08 (20140101); B60L50/64 (20190101); B60L53/80 (20190101); E05B81/18 (20140101); E05B81/28 (20140101); E05B83/00 (20140101); H01M50/249 (20210101); H01M50/262 (20210101); H01M50/271 (20210101); H01M50/574 (20210101)

U.S. Cl.:

CPC E05B81/08 (20130101); B60L50/64 (20190201); B60L53/80 (20190201); E05B81/18 (20130101); E05B81/28 (20130101); E05B83/00 (20130101); H01M50/249 (20210101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-034996 filed on Mar. 7, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an electrified vehicle that can travel by power from an electric motor that exchanges electric power with a replaceable battery.

2. Description of Related Art

[0003] Hitherto, there has been known a saddle-ride type electrified vehicle including a battery housing portion that is provided below a floor where an occupant places feet and that removably houses a battery (see, for example, Japanese Unexamined Patent Application Publication No. 2013-208962 (JP 2013-208962 A)). A lid member of the battery housing portion is integrated with the upper surface of the battery of the saddle-ride type electrified vehicle to form the floor. A power supply connector and a striker are provided on the lower surface of the battery. The battery housing portion is provided with a power receiving connector to which the power supply connector is connected, and a latch for fixing the striker. Thus, when the battery is lowered into the battery housing portion by gripping the lid member, the power supply connector is connected to the power receiving connector and the striker is fixed to the latch. When the striker and the latch are disengaged by operating a main key, the power supply connector can be detached from the power receiving connector by lifting the lid member, and the battery can be removed together with the lid member.

SUMMARY

[0004] In the related-art saddle-ride type electrified vehicle, the battery can be removed when the user disengages the striker and the latch by operating the main key. However, it is not appropriate to allow the battery to be removed in all cases from the viewpoint of safety.

[0005] In view of the above, it is a main object of the present disclosure to satisfactorily ensure safety when a battery is replaced in an electrified vehicle that can travel by power from an electric motor that exchanges electric power with the replaceable battery.

[0006] An electrified vehicle of the present disclosure includes at least one replaceable battery and an electric motor configured to exchange electric power with the battery, and is configured to travel by power from the electric motor. The electrified vehicle includes: [0007] a battery case including a case body having an opening into which the battery is to be inserted or removed, and a lid configured to open or close the opening of the case body; and [0008] a locking device configured to lock the lid in an unopenable manner at least when a start switch of the electrified vehicle is ON. [0009] In the electrified vehicle of the present disclosure, the lid of the battery case that houses the replaceable battery is locked in the unopenable manner at least when the start switch of the electrified vehicle is ON. In the electrified vehicle, there is a case where a closed circuit including the battery is or may be formed. In this case, it is possible to satisfactorily reduce the occurrence of a case where an operator who intends to replace the battery touches the battery. As a result, in the electrified vehicle of the present disclosure, it is possible to satisfactorily ensure safety when the battery is replaced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0011] FIG. 1 is a schematic configuration diagram illustrating an electrified vehicle of the present disclosure;

[0012] FIG. 2 is a schematic configuration diagram illustrating a battery case included in the disclosed electrified vehicle;

[0013] FIG. 3 is a perspective view of a battery case locking device included in the disclosed electrified vehicle;

[0014] FIG. 4 is a control diagram of the disclosed electrified vehicle; and

[0015] FIG. 5 is a time chart for explaining a welding determination process of a battery relay in electrified vehicle of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

[0016] Embodiments of the present disclosure will now be described with reference to the drawings.

[0017] FIG. 1 is a schematic configuration diagram illustrating an electrified vehicle **100** of the present disclosure. Electrified vehicle **100** shown in the drawing is a battery electric vehicle (BEV) including a plurality of replaceable batteries **10**, a system main relay SMR which is a normally open mechanical relay (reed relay), a power control device (hereinafter referred to as “PCU”) **110**, and a motor-generator MG. PCU **110** includes an inverter (drive circuitry) that drives the motor-generator MG, a boost converter, a DC/DC converter, and the like. The motor-generator MG is a synchronous generator motor (three-phase AC motor).

[0018] The rotor of the motor-generator MG is coupled to a drive shaft that rotates integrally with a drive wheel DW via a power transmission including a reduction gear and a differential gear. The motor-generator MG is driven by electric power from PCU **110** (battery **10**) and outputs a driving torque (driving force) to the drive wheel DW. In addition, the motor-generator MG outputs regenerative braking torque to the drive wheel DW at the time of braking electrified vehicle **100**, and electric power generated (regenerated) by the motor-generator is stored in the batteries **10**.

[0019] In the present embodiment, electrified vehicle **100** is a commercial vehicle used for, for example, delivery of baggage. The batteries **10** mounted on electrified vehicle **100** are replaced with fully charged batteries **10** charged by a battery charging device **80** (see FIG. 2) outside the vehicle in response to a decrease in SOC. The battery charging device **80** is installed in a collection and delivery station of a baggage, a parking facility of an electrified vehicle **100**, or the like. In addition, in a collection and delivery station or the like, the batteries **10** are taken in and out of electrified vehicle **100** or the battery charging device **80** by using the battery transport carriage **90** (see FIG. 2).

[0020] The batteries **10** each include a pack case having a substantially rectangular parallelepiped shape and a plurality of battery cells **11** accommodated in the pack case, and have the same specifications (including dimensions) as each other. The battery cell **11** of each battery **10** is, for example, a lithium ion secondary battery cell or a nickel hydrogen secondary electron cell. In the present embodiment, the battery cells **11** are connected in series in the pack case to form a cell stack. In addition, the positive terminal of the cell stack is electrically connected to the positive-electrode-side connector **12p** via a mechanical (reed) battery relay Rp. The negative terminal of the cell stack is electrically connected to the negative-electrode-side connector **12n** via a mechanical battery relay Rn.

[0021] As shown in FIG. 1, electrified vehicle **100** includes a battery case **1** in which (in the present

embodiment, for example, three) replaceable batteries **10** are accommodated in a parallel-arranged manner. The battery case **1** is fixed on the floor panel (vehicle body) of electrified vehicle **100** via a plurality of brackets **B** (see FIG. **2**) so as to be positioned behind the front seat of electrified vehicle **100**, for example. Thus, the battery accommodating portion of electrified vehicle **100** is defined. The battery case **1** includes a bottomed rectangular tubular case body **2** having an opening **2o**, and a cover **3** that opens and closes an opening **2o** of the case body **2**. In the present embodiment, the case body **2** of the battery case **1** is fixed to the floor panel such that the opening **2o** faces, for example, a side surface of electrified vehicle **100** on the driver's seat side.

[0022] Inside the case body **2** of the battery case **1**, a plurality of positive-electrode-side receptacles **4p** that can be coupled to the positive-electrode-side connector **12p** of the corresponding battery **10** and a plurality of negative-electrode-side receptacles **4n** that can be coupled to the negative-electrode-side connector **12n** of the corresponding battery **10** are disposed. The batteries **10** are connected in series via two sets of positive-side receptacles **4p** and negative-electrode-side receptacles **4n** electrically connected to each other. Further, the positive-electrode-side receptacle **4p** that is not connected to the negative-electrode-side receptacle **4n** is electrically connected to the positive-electrode-side power line **PL** via the positive-electrode-side relay of the system main relay **SMR**. The negative-electrode-side receptacle **4n** that is not connected to the positive-electrode-side receptacle **4p** is electrically connected to the negative-electrode-side power line **NL** via the negative-electrode-side relay of the system main relay **SMR**. Further, the positive-electrode-side power line **PL** and the negative-electrode-side power line **NL** are electrically connected to **PCU 110**. When the system main relay **SMR** is closed, **PCU 110** is electrically connectable to the batteries **10**. In addition, voltage sensors **V1**, **V2**, **V3** are installed in the vicinity of the battery case **1** of the floor panel of electrified vehicle **100** or in the battery case **1**. The voltage sensors **V1**, **V2**, **V3** each detect a voltage across the positive-electrode-side receptacle **4p** and the negative-electrode-side receptacle **4n** disposed in the back case of the corresponding battery **10**.

[0023] As shown in FIG. **2**, the case body **2** of the battery case **1** is formed of, for example, a plurality of metal plates that is press-processed. The case body **2** includes a bottom portion **20**, a pair of side wall portions **21** extending upward from corresponding side edges of the bottom portion **20**, a ceiling portion **22** facing the bottom portion **20** at a distance from each other, and an end wall portion **23**. The bottom portion **20**, the pair of side wall portions **21**, and the ceiling portion **22** form a flat square tube, and the end wall portion **23** closes one end portion of the square tube. As a result, a rectangular opening **2o** is defined at the end of the case body **2** that faces away from the end wall portion **23**. In addition, the positive-electrode-side receptacles **4p** and the negative-electrode-side receptacles **4n** are disposed so as to be respectively close to the end wall portion **23**. Note that the bottom portion **20** and the pair of side wall portions **21** of the case body **2** may be integrally formed, or they may be separately formed and fixed to each other.

[0024] Inside the case body **2**, (three) batteries **10** are accommodated so as to be aligned along the longitudinal direction (widthwise direction) of the opening **2o** as the inserting opening. That is, the batteries **10** are inserted into and removed from the case body **2** through the opening **2o** in the extending direction of the pair of side wall portions **21**. In addition, a plurality of partition walls **24** (see FIG. **1**) is disposed inside the case body **2** so as to be positioned between adjacent batteries **10**. Furthermore, the case body **2**, in order to be accommodated by easily positioning the battery **10** with respect to the battery case **1**, a plurality of (in this embodiment, for example, three) case-side guide portion **25** having the same height and width of each other is provided in the case body **2**.

[0025] As shown in FIG. **2**, the case-side guide portions **25** are disposed at the bottom portion **20**. That is, the case-side guide portions **25** extend from the opening **2o** side toward the end wall portion **23** side (the inside of the case body **2**) in parallel to the respective side wall portions **21**, and are spaced apart from each other in the longitudinal direction (widthwise direction) of the opening **2o**. In the present embodiment, each case-side guide portion **25** is formed by press working so as to protrude from the inner surface (upper surface) of the bottom portion **20**, which is

the inner bottom surface of the case body **2**, toward the inner side of the case body **2**, that is, toward the ceiling portion **22**. Each case-side guide portion **25** is located below the central portion in the width direction of each battery **10** accommodated in the case body **2**. Further, the upper surface of each case-side guide portion **25** is formed flat, the end portion on the opening **2o** side and the end portion on the end wall portion **23** side of each case-side guide portion **25**, an inclined surface for continuing the inner surface of the upper surface and the bottom portion **20** is formed.

[0026] The lid **3** is formed of, for example, a pressed metal plate, and is rotatably supported around the rotation axis A (see FIG. 2) by the case body **2** via a plurality of hinges (not shown). The rotation axis A extends along a lower edge of the opening **2o**, i.e., along one side of the bottom portion **20** defining the opening **2o**. As a result, the lid **3** is rotated from the top to the bottom around the rotation axis A (tilted toward the front), whereby the lid **3** can be opened to expose the opening **2o**. Further, by rotating (flipping up) the lid **3** from the bottom to the top around the rotation axis A, the lid **3** can be closed and the opening **2o** can be closed.

[0027] Further, as shown in FIG. 2, the lid **3**, so as to align with the case-side guide portion **25** corresponding when the lid **3** is opened (fully opened), a plurality of (in the present embodiment, for example, three) lid side guide portions **35** having the same height and width are disposed. The lid-side guide portions **35** are formed by press working so as to project from the inner surface of the lid **3** toward the inner side of the case body **2** and to be arranged at intervals in the longitudinal direction (width direction) of the lid **3**. The lid-side guide portions **35** are located below the central portion in the width direction of each battery **10** to be inserted and removed from the case body **2**. Further, the upper surface of each lid-side guide portion **35** is formed flat, and is included in substantially the same plane as the upper surface of the case-side guide portion **25** corresponding to when the lid **3** is opened (fully opened). Further, the end portion on the free end side of the lid **3** of the lid-side guide portion **35**, the inclined surface **35a** for continuing the inner surface of the lid **3** and the upper surface of the lid-side guide portion **35** is formed. An inclined surface **35b** for continuing the upper surface of the lid-side guide portion **35** and the inner surface of the lid **3** is formed at an end portion of the lid-side guide portion **35** on the rotation axis A side.

[0028] Each battery **10** includes a guided portion (not shown) that engages with the case-side guide portion **25** of the case body **2** and the lid-side guide portion **35** of the lid **3**. In the present embodiment, the guided portion is a pair of ridges (rails) protruding downward from the bottom surface of the battery **10** (pack case) and extending in the longitudinal direction of the battery **10**. The pair of ridges extend parallel to each other at a distance slightly larger than the width of the case-side guide portion **25** and the lid-side guide portion **35**, and are in sliding contact with the corresponding side surfaces of the case-side guide portion **25** and the lid-side guide portion **35** and the inner surface of the bottom portion **20**, respectively. Further, a contact member **15** (see FIG. 2) formed of an elastic body such as rubber or resin is provided on at least one end face of each battery **10** (pack case). The contact member **15** is disposed below the handle portion **14** of the battery **10**, and has a surface inclined so as to be separated from the end surface of the battery **10** from the upper surface side toward the bottom surface side of the battery **10**.

[0029] In addition, the case body **2** of the battery case **1** is provided with a locking device **5** that locks the lid **3** with the opening **2o** closed so as not to be opened, and two courtesy switches **6** that detect the opening and closing of the lid **3**. As shown in FIG. 3, the locking device **5** includes a locking mechanism **50** and a lock solenoid **55** that drives the locking mechanism **50** to form a locked state in response to energization. The locking mechanism **50** selectively forms an unlocked state (refer to a two-dot chain line in the drawing) that allows the lid **3** to be opened, and a locked state (refer to a solid line in the drawing) that restricts the lid **3** from being opened.

[0030] The locking mechanism **50** of the locking device **5** includes a locking lever **51** and an engaged portion **53** fixed to the lid **3**. The locking lever **51** is disposed on the end wall portion **23** side (front side in FIG. 3) of the support portion **21s** supported by the one side wall portion **21** of the case body **2** so as to extend parallel to the closed cover **3**. In addition, a central portion of the

locking lever **51** in the longitudinal direction is rotatably supported by the support portion **21s** about an axis extending in parallel to the side wall portion **21**. The engaged portion **53** is fixed to one end of the lid **3** so as to be inserted into an opening **210** formed in an outer end portion of the support portion **21s** when the lid **3** is closed (fully closed). Further, at one end (upper end) of the locking lever **51**, an engaging portion **52** engageable with the engaged portion **53** of the cover **3** inserted into the opening **210** is formed.

[0031] The lock solenoid **55** is a pull solenoid that draws a plunger connected to the shaft **57** into the coil in response to energization. The lock solenoid **55** is fixed to the end wall portion **23**-side surface of the support portion **21s** such that the shaft **57** protrudes outward from the side wall portion **21**. The shaft **57** of the lock solenoid **55** is rotatably coupled to the other end (lower end) of the locking lever **51** via a pin coupling portion. As a result, the lid **3** is closed (fully closed). At the same time, when a current is supplied to the lock solenoid **55** (coil), the engaging portion **52** of the locking lever **51** rotates toward the opening **210** in response to the shaft **57** being drawn toward the coil side, and engages with the engaged portion **53** protruding from the opening **210** toward the end wall portion **23** side. As a result, while the lock solenoid **55** is energized, the lid **3** is unlocked by the locking device **5**. It should be noted that the locking device **5** may include a push solenoid.

[0032] The two courtesy switches **6** are attached to the other side wall portion **21** of the support portion **21s** or the case body **2** so as to be able to abut the corresponding one of both end portions of the cover **3**. Each of the courtesy switches **6** is a push-type opening/closing switch that is pressed and closed by the closed lid **3** and is opened in response to opening of the lid **3**. Further, in the present embodiment, each of the courtesy switches **6** is grounded so as to form a closed circuit when closed.

[0033] The battery case **1** configured as described above is also used in the battery charging device **80**. That is, in the housing **85** of the battery charging device **80**, a plurality of battery cases **1** are arranged, for example, in the vertical direction so as to define a battery accommodating portion of the battery charging device **80** (see FIG. 2). In addition, in the present embodiment, the battery transport carriage **90** includes a battery mounting plate **95** corresponding to a case in which both of the pair of side wall portions **21** are partially removed from the bottom portion **20** of the case body **2** integrated with the pair of side wall portions **21**. The battery transport carriage **90** includes a carriage body **91** including a base frame, a plurality of casters, a handle, and the like, and a top plate **92**. The casters are at least three casters. The top plate **92** is supported by a carriage body **91** (base frame) so as to be movable up and down via an elevating mechanism (lifter). The battery mounting plate **95** is supported by the top plate **92** (the carriage body **91**).

[0034] FIG. 4 is a control diagram of electrified vehicle **100**. As shown in the drawing, electrified vehicle **100** includes an overall electronic control unit (hereinafter referred to as “BEVECU”) **200**, a motor electronic control unit (hereinafter referred to as “MGECU”) **300** for controlling PCU **110**, and a battery electronic control unit (hereinafter referred to as “battery ECU”) for managing the batteries **10**. Each of BEVECU **200**, MGECU **300** and the battery ECU **400** includes a microcomputer having a CPU, ROM, RAM, an input/output interface, and the like (not shown), various driving circuitry, various logic IC, and the like. BEVECU **200**, MGECU **300** and the battery ECU **400** exchange data (communication frames) with each other via a shared communication line (CAN bus) CB or the like.

[0035] Various sensors such as a start switch (IG switch) SS, an accelerator pedal position sensor, a shift position sensor, and a vehicle speed sensor are connected to BEVECU **200**. When electrified vehicle **100** is traveling, BEVECU **200** sets a required torque required for traveling on the basis of the accelerator operation amount and the vehicle speed. At the same time, BEVECU **200** sets a torque command or the like for the motor-generator MG based on the required torque or the like. Further, BEVECU **200** controls opening and closing of the system main relay SMR and the power supply relay (IGCT relay) **120**.

[0036] The power supply relay **120** is a normally open mechanical relay (reed relay). The power

supply relay **120** is capable of electrically connecting the auxiliary battery (low voltage battery) **130** of electrified vehicle **100**, DC/DC converters of PCU **110**, and the low voltage power line LL. The auxiliary battery **130** has a rated output-voltage of, for example, about 12 V. The low voltage power line LL is connected to a plurality of accessories including a MGECU **300**, a battery ECU **400**, and the like. A driver of the electrified vehicle **100** turns on a start switch SS, so that a start-up request of the electrified vehicle **100** is made. Then, BEVECU **200** supplies an exciting current based on electric power from the auxiliary battery **130** or the like to the coil of the power supply relay **120** to close the power supply relay **120**. By closing the power supply relay **120**, electric power from the auxiliary battery **130** or the like is supplied to the various auxiliary machines. [0037] Further, BEVECU **200** closes the power supply relay **120** and then executes predetermined processes such as the welding determination, the overheat determination, and the electric leakage determination of the system main relay SMR. When the predetermined ready-on condition is satisfied, BEVECU **200** closes the system main relay SMR. In this situation, BEVECU **200** supplies the excitation current based on the electric power from the auxiliary battery **130** or the like to the coils of the system main relay SMR (the positive-side relay and the negative-electrode-side relay) to close the system main relay SMR. When the system is started up, failures such as welding (closing failure) of the system main relay SMR may have occurred. BEVECU **200** turns on a predetermined warning light on the display provided on the instrument panel and shifts electrified vehicle **100** to the corresponding fail safe mode.

[0038] Further, BEVECU **200** shuts off the excitation current and opens the system main relay SMR when the driver turns off the start switch SS and requires electrified vehicle **100** to be shut down. Thereafter, BEVECU **200** executes predetermined processes such as the welding determination, the overheat determination, and the electric leakage determination of the system main relay SMR, and when the predetermined conditions are satisfied, shuts off the supplying of the excitation current and opens the power supply relay **120**. If a fail occurs, such as welding of the system main relay SMR, when system shutdown is requested, BEVECU **200** turns on the predetermined fail flag and then shuts down electrified vehicle **100**. Electrified vehicle **100** then transitions to fail safe mode when the start switch SS is turned on next time. In electrified vehicle **100**, the display of the instrument panel is turned off when the start switch SS is turned off.

[0039] As illustrated in FIG. 4, the battery ECU **400** acquires detected values of the voltage sensors V1, V2, V3 provided for each of the batteries **10**. Further, the battery ECU **400** is electrically connected to the battery relays Rp, Rn of the respective batteries **10** via connectors and receptacles (not shown) and a power supply line. While the power supply relay **120** is closed, the battery ECU **400** supplies an exciting current to the coils of each battery relay Rp, Rn to close each battery relay Rp, Rn. The excitation current is based on electric power supplied from the auxiliary battery **130** or the like via the power supply relay **120**. Further, the battery ECU **400** may cut off the energization current to open the respective battery relays Rp, Rn. When at least one of the battery relays Rp, Rn is opened, even if the system main relay SMR is closed, the batteries **10**, a PCU **110**, a motor-generator MG, and the like are included (closed circuit).

[0040] Further, the battery ECU **400**, after the power supply relay **120** is closed in response to the start switch SS is turned on (and before the system main relay SMR is closed), and after the system main relay SMR is opened in response to the start switch SS being turned off, to determine whether or not there is welding (closing failure) of the battery relays Rp, Rn of the respective batteries **10**. That is, when the start switch SS is turned on and the power supply relay **120** is closed, the battery ECU **400** determines whether one of the battery relays Rp, Rn (for example, the battery relay Rp) of the respective batteries **10** is welded, as illustrated in FIG. 5. Note that the relays Rp₁ and Rn₁ in FIG. 5 indicate the battery relays Rp, Rn of the first battery **10**. The relays Rp₂ and Rn₂ indicate the battery relays Rp, Rn of the second battery **10**. The relays Rp₃ and Rn₃ indicate the battery relays Rp, Rn of the third battery **10**.

[0041] Specifically, when the power supply relay **120** is closed, the battery ECU **400** closes only

the other of the battery relays Rp, Rn (for example, the battery relay Rn) of the respective batteries **10** (time to in FIG. 5). The battery ECU **400** determines whether one of the battery relays Rp, Rn of the respective batteries **10** is welded based on the detected value of the voltage sensors V1, V2, V3. When the detected values of the voltage sensors V1, V2, V3 are all zero, the battery ECU **400** as the failure determination device determines that one of the battery relays Rp, Rn of each battery **10** is not welded, and closes one of the battery relays Rp, Rn of each battery **10**. At the same time, the battery ECU **400** transmits a notification of permission to close the system main relay SMR to BEVECU **200** (time t1 in FIG. 5). In this case, BEVECU **200** closes the system main relay SMR on condition that the system main relay SMR or the like is normal (time t2 in FIG. 5). On the other hand, the battery ECU **400** may determine that at least one of the battery relays Rp, Rn of the respective batteries **10** is welded based on the detected value of the voltage sensors V1, V2, V3. In this case, the battery ECU **400** turns on a predetermined warning light on the display provided on the instrument panel and shifts electrified vehicle **100** to the corresponding fail safe mode.

[0042] Further, when the start switch SS is turned off and the system main relay SMR is opened (time t3 in FIG. 5), the battery ECU **400** determines whether or not the other of the battery relays Rp, Rn (for example, the battery relay Rn) of the respective batteries **10** is welded. Specifically, when the system main relay SMR is opened, the battery ECU **400** opens only the other of the battery relays Rp, Rn of the respective batteries **10** (time t4 in FIG. 5). The battery ECU **400** determines whether the other of the battery relays Rp, Rn of the respective batteries **10** is welded based on the detected values of the voltage sensors V1, V2, V3. When the detected values of the voltage sensors V1, V2, V3 are all zero, the battery ECU **400** as the failure determination device determines that the other of the battery relays Rp, Rn of the respective batteries **10** is not welded. The battery ECU **400** opens one of the battery relays Rp, Rn of the respective batteries **10** and transmits a notification of opening permission of the power supply relay **120** to BEVECU **200** (time t5 in FIG. 5). Upon receiving the opening permission notification from the battery ECU **400**, BEVECU **200** opens the power supply relay **120** on condition that the system main relay SMR or the like is normal.

[0043] Further, as shown in FIG. 4, the lock solenoid **55** (coil) of the locking device **5** of the battery case **1** is connected to the low voltage power line LL. While the power supply relay **120** is closed, a current from the auxiliary battery **130** or the like is constantly supplied to the lock solenoid **55**. As a result, while the power supply relay **120** is closed, the lock solenoid **55** holds the locking lever **51** of the locking mechanism **50** so that the engaging portion **52** is engaged with the engaged portion **53** inserted into the opening **210**. As a consequence, the cover **3** of the battery case **1** is unlocked by the locking device **5** while the start switch SS is turned on and the power supply relay **120** is closed.

[0044] Furthermore, as shown in FIG. 4, the two courtesy switches **6** of the battery case **1** are electrically connected to one end of a coil **150c** of an interlock relay (circuit opening and closing relay) **150** via a connector and a receptacle (not shown) and an electric wire. The interlock relay **150** is a mechanical relay (reed relay). The other end of the coil **150c** of the interlock relay **150** is connected to the low voltage power line LL. One of the two contacts of the interlock relay **150** is electrically connected to the first terminal of BEVECU **200**, and the other of the two contacts is electrically connected to the second terminal of BEVECU **200**. BEVECU **200** applies a voltage based on electric power from the auxiliary battery **130** or the like to the first terminal during a period from when the start switch SS is turned on to when it is turned off.

[0045] When the cover **3** of the battery case **1** is closed and at least one of the two courtesy switches **6** is closed, one end of the coil **150c** of the interlock relay **150** is grounded by the courtesy switch **6**. Therefore, when the cover **3** of the battery case **1** is closed and the power supply relay **120** is closed, an excitation current based on electric power from the auxiliary battery **130** or the like is supplied to the coil **150c**, thereby closing the interlock relay **150**. When the interlock relay **150** is closed, the first and second terminals of BEVECU **200** are electrically connected via the

interlock relay **150**. BEVECU **200** compares the voltages of the first and second terminals and determines that the interlock relay **150** is closed if the voltages of the first and second terminals are approximately the same.

[0046] On the other hand, when the cover **3** of the battery case **1** is opened, even if the power supply relay **120** is closed, the energization current to the coil **150c** is interrupted by the opening of the respective courtesy switches **6**, and the interlock relay **150** is opened. The opening of each courtesy switch **6** is the opening of the closed circuit by releasing the ground. Then, the first and second terminals of BEVECU **200** are disconnected from each other, and the voltage of the second terminal becomes zero. BEVECU **200** determines that the lid **3** of the battery case **1** is opened when the voltage of the second terminal drops (drops to zero), and the respective courtesy switches **6** are opened to open the interlock relay **150**. BEVECU **200** cuts off supplying the excitation current to the system main relay SMR to open the system main relay SMR. As a result, a high-voltage circuit (closed circuit) including the batteries **10**, PCU **110**, the motor-generator MG, and the like is opened.

[0047] In addition, as shown in FIG. **4**, a light emitter **140** as a notification device is electrically connected to the battery ECU **400**. The light emitter **140** includes a light emitting diode or the like. The light emitter **140** is installed in front of the lid **3** so as to be identified by an operator (user) who replaces the battery **10** in a state where the lid **3** of the battery case **1** is closed (see FIG. **2**). The front of the cover **3** is in the vicinity of the battery case **1** of electrified vehicle **100** floor panel. However, the light emitter **140** may be installed in the battery case **1** as long as it can be identified by an operator in a state where the lid **3** of the battery case **1** is closed. The battery ECU **400** supplies an exciting current to the light emitter **140** to turn on (emit light) the light emitter **140** while the start switch SS is turned on and the power supply relay **120** is closed. The excitation current is based on electric power supplied from the auxiliary battery **130** or the like via the power supply relay **120**.

[0048] That is, in electrified vehicle **100**, the cover **3** of the battery case **1** is unlocked by the locking device **5** while the start switch SS is turned on and the power supply relay **120** is closed. At the same time, the light emitter **140** installed in the vicinity of the battery case **1** is made to emit light. Therefore, the light emitter **140** functions as a notification device that notifies that the lid **3** of the battery case **1** is locked. In addition, the battery ECU **400** can blink the light emitter **140** by intermittently supplying the exciting current.

[0049] As described above, electrified vehicle **100** includes the replaceable batteries **10** and a motor-generator MG that exchanges power with the batteries **10**. Electrified vehicle **100** can be driven by power from the motor-generator MG. Further, electrified vehicle **100** includes a battery case **1** and a locking device **5**. The battery case **1** includes a case body **2** having an opening **2o** into which the batteries **10** are inserted and removed, and a cover **3** that opens and closes an opening **2o** of the case body **2**. The locking device **5** locks the lid **3** in a non-openable manner at least when electrified vehicle **100** starting-switch SS is switched on.

[0050] More specifically, the cover **3** of the battery case **1** is locked so as not to be opened while electrified vehicle **100** start switch SS is turned on and the power supply relay **120** is closed. Thus, in electrified vehicle **100**, when a high-voltage circuit (closed circuit) including a plurality of batteries **10**, a PCU **110**, a motor-generator MG, and the like is formed, or when there is a possibility that the high-voltage circuit is formed after the start switch SS is turned on, it is possible to satisfactorily suppress the operator who intends to replace the battery **10** from touching the battery **10**. Consequently, electrified vehicle **100** can ensure good safety when replacing the battery **10**.

[0051] The locking device **5** of the battery case **1** includes a locking mechanism **50** and a lock solenoid **55**. The locking mechanism **50** selectively forms an unlocked state in which the opening of the lid **3** is permitted and a locked state in which the opening of the lid **3** is restricted. The lock solenoid **55** drives the locking mechanism **50** so as to form a locked state in response to

energization, and the lock solenoid **55** of the locking device **5** is connected to a power source such as the auxiliary battery **130** via a power supply relay **120** that is closed in response to the start switch **SS** being turned on. Accordingly, the locking device **5** can be reliably operated when the start switch **SS** is turned on while the control in electrified vehicle **100** is suppressed from becoming complicated.

[0052] Further, in electrified vehicle **100**, the lock solenoid **55** is supported by the case body **2** of the battery case **1**. Further, the locking mechanism **50** includes a locking lever **51** rotatably supported by the case body **2**, and the locking lever **51** is rotationally driven by the lock solenoid **55** to engage with the engaged portion **53** provided on the lid **3**. Accordingly, the locking device **5** can be integrated with the battery case **1** to further improve the assemblability and versatility of the battery case **1** to electrified vehicle **100**. However, the locking device **5** may be installed on an electrified vehicle **100** vehicle body such as a floor panel.

[0053] Further, the battery case **1** of electrified vehicle **100** includes a plurality of courtesy switches **6** for detecting opening and closing of the cover **3**. The respective courtesy switches **6** are closed when the cover **3** is closed to allow the high-voltage circuit (closed circuit) including the batteries **10**, PCU **110**, the motor-generator **MG**, and the like to be closed. At the same time, each of the courtesy switches **6** is opened when the lid **3** is open, and the closing of the high voltage circuit is prohibited. Accordingly, when the lid **3** of the battery case **1** is opened, a high voltage circuit through which a current flows from the battery **10** is not formed, and thus safety at the time of replacement of the battery **10** can be more satisfactorily ensured. However, the battery case **1** may include a single courtesy switch **6**.

[0054] Further, electrified vehicle **100** includes an interlock relay (circuit opening and closing relay) **150** that opens and closes a high-voltage circuit (closed circuit) including a plurality of batteries **10**, a PCU **110**, a motor-generator **MG**, and the like in cooperation with a BEVECU **200**. The high-voltage circuitry includes the batteries **10**, PCU **110**, motor-generator **MG**, and the like. That is, the interlock relay **150** closes when the exciting current is supplied to the coil **150c** to allow the high voltage circuit to be closed, and opens when the exciting current is not supplied to the coil **150c** to prohibit the high voltage circuit from being closed. In addition, the respective courtesy switches **6** of the battery case **1** are closed when the cover **3** is closed to allow the exciting current to be supplied to the interlock relay **150** (coil **150c**). At the same time, each of the courtesy switches **6** opens when the lid **3** of the battery case **1** is opened, thereby interrupting the supply of the excitation current to the interlock relay **150**. Thus, when the lid **3** of the battery case **1** is open, it is possible to more reliably suppress the formation of a high voltage circuit through which a current flows from the batteries **10**.

[0055] In the above-described embodiment, the interlock relay **150** has a function of notifying BEVECU **200** of the opening/closing status of the cover **3** of the battery case **1**, but the present disclosure is not limited thereto. That is, the interlock relay **150** may, for example, interrupt the supplying of the excitation current to the system main relay **SMR** when the courtesy switch **6** is opened. In addition, instead of installing the voltage sensors **V1**, **V2**, **V3** in electrified vehicle **100** or the battery case **1** described above, a voltage sensor **V1**, **V2** or **V3** may be provided. The voltage sensor **V1**, **V2** or **V3** detects a voltage across the positive-electrode-side receptacle **4p** and the negative-electrode-side receptacle **4n** in the pack case of the corresponding battery **10**. Further, a capacitor may be provided between the positive-electrode-side receptacle **4p** and the negative-electrode-side receptacle **4n** of the respective batteries **10**, and a single voltage-sensor may be provided in electrified vehicle **100** or the battery case **1**. Further, the batteries **10** may be provided with indicator lights that are turned on when the battery relays **Rp** or **Rn** are closed. Further, the battery case **1** may be mounted on a moving object other than an electrified vehicle **100** such as a railcar, or may be installed in a fixed facility other than the battery charging device **80**. The battery case **1** may also be configured to accommodate a single replaceable battery **10**. Furthermore, the case body **2** of the battery case **1** may be defined by an electrified vehicle **100** vehicle body. A part

of the case body 2 may be formed by a part of a vehicle body of electrified vehicle 100.

[0056] It is needless to say that the disclosure of the present disclosure is not limited to the above-described embodiments, and various modifications can be made within the scope of the extension of the present disclosure. Furthermore, the above-described embodiment is only a specific form of the disclosure described in the column of the outline of the disclosure, and does not limit the elements of the disclosure described in the column of the outline of the disclosure.

[0057] The present disclosure is applicable to the manufacturing industry of electrified vehicle and the like.

Claims

1. An electrified vehicle that includes at least one replaceable battery and an electric motor configured to exchange electric power with the battery, and that is configured to travel by power from the electric motor, the electrified vehicle comprising: a battery case including a case body having an opening into which the battery is to be inserted or removed, and a lid configured to open or close the opening of the case body; and a locking device configured to lock the lid in an unopenable manner at least when a start switch of the electrified vehicle is ON.
 2. The electrified vehicle according to claim 1, wherein: the locking device includes a locking mechanism configured to selectively set an unlocked state in which the lid is allowed to be opened and a locked state in which opening of the lid is restricted, and a solenoid configured to drive the locking mechanism to set the locked state in response to energization; and the solenoid of the locking device is connected to a power supply via a relay that is closed when the start switch is turned ON.
 3. The electrified vehicle according to claim 2, wherein: the solenoid is supported by the case body; and the locking mechanism includes a locking lever that is rotatably supported by the case body and rotationally driven by the solenoid to engage with an engaged portion provided on the lid.
 4. The electrified vehicle according to claim 1, wherein: the battery case includes a switch configured to detect opening and closing of the lid; and the switch is configured to allow closing of a closed circuit including the battery when the lid is closed, and prohibit the closing of the closed circuit when the lid is open.
 5. The electrified vehicle according to claim 4, further comprising a circuit opening and closing relay configured to allow the closing of the closed circuit by being closed when an excitation current is supplied, and prohibit the closing of the closed circuit by being opened when the excitation current is not supplied, wherein the switch is configured to allow supply of the excitation current to the circuit opening and closing relay by being closed when the lid is closed, and interrupt the supply of the excitation current to the circuit opening and closing relay by being opened when the lid is opened.
-